

2	(a) (i) amplitude scale reading 2.2 (cm) amplitude = $2.2 \times 2.5 = 5.5 \text{ mV}$	C1 A1	[2]
	(ii) time period scale reading = 3.8 (cm) time period = $3.8 \times 0.5 \times 10^{-3} = 0.0019 \text{ (s)}$	C1 C1	
	frequency $f = 1 / 0.0019 = 530 \text{ (526) Hz}$	A1	[3]
	(iii) uncertainty in reading = ± 0.2 in 3.8 (cm) or 5.3% or 0.2 in 7.6 (cm) or 2.6% [allow other variations of the distance on the x-axis]	M1	
	actual uncertainty = 5.3% of 526 = 27.7 or 28 Hz or 2.6% of 526 = 13 or 14	A1	[2]
	(b) frequency = $530 \pm 30 \text{ Hz}$ or $530 \pm 10 \text{ Hz}$	A1	[1]